Environmental Data Analysis — New Practice Exercises

Exercise 1 — Statistical Populations and Sampling Units

For each of the following research questions:

1. Define an appropriate **statistical population**.
2. Identify the **experimental/observational unit**.
3. Suggest one possible method for collecting the data.

a. What is the relationship between annual rainfall and crop yield for maize farms in southwestern Nigeria?

b. Does the amount of urban tree cover influence average summer temperature in major African cities?

c. What environmental factors affect fish abundance in coastal lagoons of West Africa?

d. How does fertilizer application influence nitrate concentration in nearby groundwater wells?

e. What is the relationship between traffic density and airborne particulate matter (PM2.5) near schools in urban areas?

Exercise 2 — Identifying Data Types

For each study below:

- Determine the **type of data** (continuous, binary, count, proportion, time series, categorical, circular, etc.).
- Identify the **dependent variable**.
- Identify the **independent variable(s)**.
- Define the **sampling unit**.
- Define the **statistical population**.
- State the **sample size**.

a. Daily river discharge measurements were collected for 730 consecutive days from a watershed in Ghana.

b. Researchers recorded whether or not 250 households recycled plastic waste during a one-month period.

c. Soil moisture was measured every 5 meters along a 2 km transect in a forest reserve.

d. Monthly mosquito counts were recorded at 15 wetlands over two years.

e. Wind direction was recorded every hour at a coastal weather station for six months.

Exercise 3 — Environmental Data Exploration

Obtain a real environmental dataset from any public source (for example: climate, air pollution, soil, vegetation, rainfall, or biodiversity data).

Answer the following:

1. What is the source of the dataset?
2. What variables are included?
3. Which variables are quantitative and which are qualitative?
4. Are there missing values?
5. What type of statistical analyses might be appropriate for this dataset?

R Programming Exercises

Exercise 4 — Basic Mathematical Operations

Assign:

```
x <- 5
y <- 8
```

Compute the following:

1. ((xy + 4))
2. (+ y^2)
3. (_{10}())
4. (e^{x-y} + 5^x)

Exercise 5 — Working with Vectors

Consider the vector:

```
x <- c(12, 5, 8, 15, 7, 20, 11, 3)
```

1. Which values are greater than 10?
2. Which values are equal to 7?
3. Calculate the median of the vector.
4. Calculate the standard deviation.

5. Create a subvector containing values less than the mean.
6. Find the minimum value in the subvector.

Exercise 6 — Matrix Operations

Create the matrix:

```
A <- matrix(c(2,4,6,8,1,3,5,7), nrow = 2, byrow = TRUE)
```

1. Assign column names to the matrix.
2. Multiply every element by 2 and store the result in matrix `B`.
3. Subtract `A` from `B`.
4. Add a third row `c(9,10,11,12)` to the matrix using `rbind()`.
5. Extract the second column of the new matrix.

Exercise 7 — Character Vectors and Strings

1. Use `LETTERS` to obtain the 12th letter of the alphabet.
2. Use code to obtain the first letter of the alphabet.
3. Create the vectors:

```
students <- c("Mary", "James", "Peter")
courses <- c("Biology", "Statistics", "Geography")
```

4. Combine the vectors into a table using `cbind()`.
5. Use `paste()` to create sentences like:

```
"Mary studies Biology"
```

Exercise 8 — Data Frames

Create the following vectors:

```
city <- c("Lagos", "Abuja", "Ibadan")
rainfall <- c(120, 98, 110)
flood <- c(TRUE, FALSE, TRUE)
```

1. Create a data frame named `weather_df`.
2. Display the structure of the data frame.
3. Add a new column called `temperature`.

4. Add a new row of observations.
5. Produce a summary of the updated data frame.

Exercise 9 — Built-in Dataset Practice

Use the built-in dataset `mtcars`.

1. Display the first 10 rows.
2. Rename the variable `mpg` to `mileage`.
3. Create a new variable converting mileage from miles per gallon to kilometers per liter.
4. Identify the cars with mileage above the average.
5. How many cars have more than 6 cylinders?
6. Which car has the highest horsepower?

Exercise 10 — Missing Values and Dates

Import a CSV dataset named:

```
daily_temperature.csv
```

1. Replace all values equal to `-999` with `NA`.
2. Convert the `DATE` column to class `Date`.
3. Count the number of missing observations.
4. Remove rows containing missing values.
5. Create a plot of temperature versus date.